

StreamSense: Viewer Engagement & Retention Insights

University Capstone Group Project

1. Problem Statement

A subscription-based streaming platform already captures rich viewer interaction data — watch duration, pause frequency, and episode completion — for every viewing session. The problem is that this data sits unused when it matters most: the content acquisition team still greenlights new shows and renews existing ones based on gut feel and headline numbers like total views, without any understanding of which engagement patterns actually predict whether a viewer sticks around.

That disconnect is costly. Two shows can post similar view counts on paper, but one might be watched in full with almost no pausing while the other is abandoned halfway through by most viewers. Without connecting engagement behaviour to retention outcomes, the acquisition team has no reliable way to tell which of those two shows is actually worth renewing — they are, in effect, making million-dollar content bets blind.

2. Objectives

Our goal for this project is to give the acquisition team an evidence-based way to judge content, not to redesign the streaming product itself. Specifically, we want to:

- Bring watch duration, pause frequency, and completion data together into one clean, analysis-ready dataset
- Build an engagement score that reflects how genuinely absorbed viewers are in a piece of content, not just raw watch time
- Test whether that engagement score actually correlates with retention outcomes such as renewal, using historical data
- Give the acquisition team a simple, ranked view of which content is performing well versus poorly on engagement and retention
- Make sure every score and recommendation is explainable, so it can be defended in a real content decision, not treated as a black box

3. Stakeholders

We grouped stakeholders into four categories, since who would use this tool and who benefits from it are not the same people.

3.1 Primary Users

These are the people who would actually open this tool to make a decision.

Stakeholder	Why they're primary
Content Acquisition Analyst	They decide which shows to renew, drop, or license next — this tool is built to inform that call directly

3.2 Secondary Users

These people use the output of the system, but it isn't their main day-to-day tool.

Stakeholder	How they use it
-------------	-----------------

Content Strategy Lead	Reviews engagement-vs-retention trends across genres to guide future content buying strategy
Marketing Team	Checks which shows have high engagement but low awareness, to target promotion where it can move the needle
Data / BI Team	Maintains the underlying pipeline and validates that new data keeps joining correctly

3.3 Decision Makers

This is who signs off on whether the scoring approach is sound and worth acting on.

Stakeholder	Their role
Head of Content Acquisition	Approves the engagement scoring logic and decides whether it factors into real renewal decisions

3.4 Business Beneficiaries

These aren't people who touch the tool at all — they benefit from the outcome.

Stakeholder	Why they benefit
Company Leadership	Fewer poor content bets means content budget is spent on titles that actually retain subscribers
Subscribers (indirectly)	A catalogue shaped by what genuinely keeps people engaged, rather than by guesswork

4. Functional Requirements

What the system needs to actually do:

- Ingest and clean session-level viewing data — watch duration, pause count, and completion percentage per user per title
- Calculate a per-title engagement score that combines completion rate, watch duration, and pause behaviour
- Join engagement scores against historical retention/renewal outcomes to test which engagement patterns actually correlate with a viewer staying subscribed
- Rank titles by engagement score and by their observed link to retention, so analysts can see standout and underperforming content at a glance
- Let users filter results by genre, subscription tier, or time window
- Show the breakdown behind each score — not just a single number with no explanation
- Flag titles where engagement and completion look unusually low relative to similar content, for closer review
- Produce a dashboard view and a simple exportable summary report for use in acquisition meetings
- Run basic data quality checks (missing values, invalid durations, duplicate sessions) each time the dataset is refreshed

5. Success Metrics (KPIs)

How we'll know this actually worked, not just that it runs:

- The engagement score meaningfully separates titles with high historical renewal rates from those with low renewal rates in our test data
- At least 95% of viewing session records join correctly across the engagement dataset fields we rely on
- An analyst can identify the top and bottom performing titles for a given period without needing us to explain the dashboard first
- The reasoning behind any single title's score can be explained in plain language during a walkthrough or demo
- The final report clearly states which engagement patterns (e.g. high pause frequency, low completion) are associated with churn versus retention

6. Scope

What we're building for this project

- Cleaning and preparing the viewer engagement dataset (watch duration, pause frequency, completion %)
- A working engagement score formula with reasoning we can defend
- Analysis correlating engagement patterns with historical retention/renewal outcomes
- A ranked, filterable view of content performance
- A simple dashboard (e.g. Streamlit or Power BI) for walkthroughs and demos

What we're intentionally leaving out for now

- Live/real-time data processing — we're working with a static historical dataset for this project
- A full churn-prediction machine learning model — treated as a possible future extension, not core to this deliverable
- Automated content renewal or removal decisions — this tool informs the acquisition team, it doesn't decide for them
- A production-grade or mobile version of the dashboard

7. Assumptions

- We'll have access to a viewing session dataset (watch duration, pause frequency, completion %) in a usable, exportable format
- The dataset includes, or can be paired with, historical renewal/churn outcomes so we can validate the engagement score against real results
- There's enough historical data to meaningfully test whether engagement patterns correlate with retention
- The dataset can be joined on common fields such as User_ID and Content_ID — this is the first thing we're confirming before locking in our approach
- This is a course/capstone deliverable, so the dashboard and analysis are evaluated on clarity and reasoning rather than production scalability